Methodology

***RQ1* - What key factors lead to late deliveries, and how can we proactively identify high-risk orders to improve customer satisfaction?**

Research Objective:
This research will aim to identify the most critical factors that lead to late deliveries on the Olist e-commerce platform and develop a predictive model that flags high-risk orders before they are delivered. The objective is to enable proactive interventions that reduce delivery failures and improve customer satisfaction.

Data Preparation:
The analysis will use the Olist Combined Dataset, which merges transactional, product, customer, review, seller, and payment data into a comprehensive record of each orderResearch Questions. The relevant attributes will include:

- **Timestamps:**
  - order_purchase_timestamp
  - order_delivered_customer_date
  - order_estimated_delivery_date
  - shipping_limit_date

- **Customer Feedback:**
  - review_score

- **Product Details:**
  - product_category_name
  - product_weight_g
  - product_volume_cm3

- **Payment Details:**
  - payment_type
  - payment_installments
  - payment_value

- **Demographics:**
  - customer_state
  - seller_state

A binary target variable late_delivery will be created using:
late_delivery = 1 if order_delivered_customer_date > order_estimated_delivery_date else 0

Derived features will include:
- days_to_ship = days between purchase and shipping deadline
- delivery_time = days from purchase to actual delivery
- estimated_delivery_days = days from purchase to estimated delivery

- review_flag = 1 if review score ≤ 3, otherwise 0

Geospatial coordinates and mapping will not be used in this analysis.

Feature Encoding and Cleaning:
Categorical variables (e.g., payment_type, customer_state, product_category_name) will be encoded using one-hot encoding. Numerical features (e.g., product weight, payment value) will be scaled using Min-Max Normalization to ensure consistent feature ranges.

Missing values will be imputed using median (for numeric features) or mode (for categorical features). Outliers in features like product dimensions and delivery time will be handled using the Interquartile Range (IQR) method.

Modeling Techniques:
To classify whether an order will be delivered late, several supervised classification algorithms will be implemented. These models will be compared to identify the most effective approach for early detection of high-risk orders:

- **Logistic Regression:** Will serve as a baseline model to evaluate how linear changes in features influence the odds of late delivery. Its coefficients will also offer direct interpretability for operational use.

- **Random Forest Classifier:** Will be used to model complex interactions and non-linear relationships. It will provide feature importance rankings, helping to identify which variables (e.g., days_to_ship, product_volume_cm3) are most predictive of delays.

- **XGBoost (Extreme Gradient Boosting):** Will be employed for its strength in optimizing classification problems with imbalanced data. It improves upon decision trees through iterative boosting and will likely offer strong performance.

- **Support Vector Machine (SVM):** Will be tested to determine how well it separates on-time and late delivery classes in a high-dimensional space. Though less interpretable, it serves as a useful benchmark model.

- **Artificial Neural Network (ANN):** An ANN model will be constructed to capture complex non-linear relationships between features. The network will include:
    - One input layer (equal to the number of input features after encoding)
    - One or more hidden layers using ReLU activation
    - One output layer using a sigmoid activation for binary classification

  The ANN will be trained using the Adam optimizer and binary cross-entropy loss.

Performance will be evaluated using accuracy, precision, recall, F1-score, and AUC. Regularization techniques such as dropout and early stopping will be applied to prevent overfitting.

Model Evaluation:

All models will be evaluated based on:

- Precision, Recall, and F1-score, with emphasis on Recall to reduce false negatives
- ROC-AUC to assess class separation quality
- Confusion Matrix for detailed error analysis

To address the class imbalance between late and on-time deliveries, the following techniques will be used:

- SMOTE (Synthetic Minority Over-sampling Technique) to balance the training set
- Class weight adjustments to penalize misclassification of minority class instances

Optimization and Thresholding:
Hyperparameters for each model will be tuned using Grid Search CV. For the ANN, architecture tuning will involve varying the number of layers, neurons, and dropout rates based on validation performance.

Prediction thresholds will be calibrated based on business objectives — for instance, flagging an order as high-risk even at lower predicted probabilities if the cost of a missed delay is high.

Interpretation and Business Use:
Feature importance outputs from Random Forest, XGBoost, and ANN will be used to identify the key drivers of delivery delays. These insights will enable Olist to:

- Automatically flag high-risk orders for preemptive action
- Inform dynamic adjustments to estimated delivery windows
- Improve seller management and inventory planning for high-risk categories or delivery timelines
- Increase customer satisfaction by aligning expectations with historical and real-time fulfillment data

This methodology will provide a robust, scalable framework for improving delivery performance across Olist's operations.

***RQ2* - *Which sellers consistently delay order handoffs to carriers, and what underlying patterns or behaviors contribute to this? Can early seller delays help us anticipate customer-facing delivery issues*?**

(Note: geospatial features will not be used as part of this analysis as we do not have the computing power necessary to calculate distance values based on latitude and longitude coordinates over such a large dataset.)

To answer either of the parts of this question, we first need to dig further into the definition of a seller handoff delay. The exploratory data analysis (EDA) showed that defining handoff delay as the difference between order/payment approval and the date the seller delivered the ordered item to the carrier resulted in a significant number of orders (thousands based on visual estimation from the histogram in section VII.A of the EDA document)[1] where that value was actually negative, which should be impossible, since the seller would presumably not have information that an order was placed before payment was made. It's possible that the differences in time zones across Brazil (there are 4 time zones across Brazil, much like the continental United States) would make it possible to have a payment approved within an hour or two after midnight in an easterly time zone and then the order delivered the "previous day" to a carrier to the west, but even in that scenario, a value that's more negative than -1 should not be possible. It's possible that Olist does not require payment (or a payment plan) to be approved/cleared before informing the seller of a shipping requirement, though we do not have that information at hand. Therefore, the in-depth analysis of seller delay and its relation to late customer deliveries will proceed as follows:

1. Create a handoff_delay_alt field (in days, as with primary handoff_delay) as the difference between order placement (order_purchase_timestamp) and delivery to the carrier
2. Review handoff_delay_alt descriptive statistics and plot histogram, then compare to primary handoff_delay statistics and histogram to determine which is more appropriate for further analysis
3. Eliminate negative values with absolute value >1 from chosen delay field, since for negative values to be real would violate the laws of physics.
    a. Convert all -1 values to 0 based on the assumption that it is impossible to deliver an order for shipment before that order is placed, and that all -1 delay times represent handoffs performed in positive time that are negative only due to time zone effects.
4. Create promised_handoff_delay field as the difference between the shipping_limit_date field (date determined by system on which the handoff is due) and order_delivered_carrier_date field
5. Perform correlation analysis between promised_handoff_delay and chosen handoff delay field
    a. Use Spearman correlation due to better performance with continuous, non-normal data
6. Create 'delivery_delay' field as difference between order_estimated_delivery_date and order_delivered_customer_date
7. Perform correlation analysis between chosen delay field and delivery_delay
    a. Use Spearman correlation as in step 5(a)
8. The EDA showed that 7.6% of all orders were delivered late (~8900/115k), so create a parallel handoff_late field to indicate True if order_delivered_carrier_date > shipping_limit_date
    a. Display descriptive statistics and value counts for True/False
9. Determine number of cases where both delivered_late and handoff_late are True
    a. Visualize via Venn diagram the intersection between delivered_late and handoff_late as part of all orders
10. Create bar chart of sellers in descending order of number of combined handoff_late = True & delivered_late = True instances, with cumulative count line to show where cutoff for 80% of combined instances occurred (Pareto Principle)
11. Determine top 10 product categories by 'both_late' (delivered_late == handoff_late == True), similar to plot shown in EDA Section VI.B.

If sufficient evidence is discovered in Steps 1-11 that there is enough information to possibly predict cases where late handoffs possibly led to late deliveries, and that the seller is a critical category for that determination (since sellers are the primary focus of the question), then the following models will be created to attempt to predict late deliveries that will involve late handoff:

1. Logistic Regression - This model is included to permit use of both numeric and categorical variables to predict probability of concurrent late handoff and late deliveries. Optimization via feature selection and cross-validation will isolate the best features as measured by recall_score, since we look to minimize the number of false negatives.
2. Support Vector Machine - This model is included to compare a non-parametric predictor with the logistic regression, which is a parametric predictor. We will specifically be using the linear support vector classifier model from scikit-learn since it is friendlier to large datasets. Optimization will be performed via feature selection (forward- and backward-sequential) and hyperparameter tuning of the C and tolerance parameters of the model.

The best model will be chosen using the best combination of precision score and recall score from the two, recognizing that we want to minimize false negatives before minimizing false positives.

<u>RQ3</u> - How do customer segments—such as first-time vs. repeat buyers or by region—differ in cancellation and return behavior, and can we predict these actions early based on browsing or shopping patterns to reduce churn?

Research Objective:
This study aims to analyze differences in order cancellation and product return behaviors across distinct customer segments, specifically, first-time versus repeat buyers, and by geographic region, and to develop predictive models utilizing early-stage browsing and shopping activity indicators. The goal is to proactively identify customers at higher risk of cancellation or return, enabling targeted interventions to reduce customer churn and improve overall satisfaction.

Data Preparation:
The analysis will use the Olist Combined Dataset, which merges transactional, product, customer, review, seller, and payment data into a comprehensive record of each order Research Questions. Below are those attributes that will be used for the analysis by groups:

Customer Characteristics (Identifying Customer Type and Region)

- **customer_state:** Customer's state
- **customer_city:** Customer's city

Order Time Information (Analyzing Customer Purchasing Behavior)

- **order_purchase_timestamp:** Time when the order was placed
- **order_delivered_carrier_date:** Time when the order was handed over to the carrier
- **order_estimated_delivery_date:** Estimated delivery date

- **order_delivered_customer_date:** Actual delivery date to the customer
- **shipping_limit_date:** Deadline for shipping

Product Information (Potentially Influencing Returns and Cancellations)

- **product_category_name_english:** Product category (English)
- **product_weight_g:** Product weight in grams
- **product_length_cm:** Product length in centimeters
- **product_height_cm:** Product height in centimeters
- **product_width_cm:** Product width in centimeters
- **price:** Product price
- **freight_value:** Shipping cost

Payment Information (Potentially Related to Cancellations)

- **payment_type:** Payment method
- **payment_installments:** Number of payment installments
- **payment_value:** Payment amount
- **payment_sequential:** Payment sequence

Customer Reviews (Potential Indicators of Customer Satisfaction and Likelihood of Returns)

- **review_score:** Review rating score
- **review_comment_message:** Review message content
- **review_creation_date:** Review creation date
- **review_answer_timestamp:** Seller's response timestamp to the review

Seller Information (Geography Might Influence Product Returns)

- **seller_state:** Seller's state
- **seller_city:** Seller's city

Feature Encoding and Data Cleaning:

Categorical variables such as payment_type, customer_state, and product_category_name will be transformed using one-hot encoding to ensure effective representation for modeling. Numerical features like product_weight and payment_value will be standardized using Min-Max normalization to maintain consistent feature scales.

Missing values in numerical variables will be imputed using the median, while categorical variables will be filled using the mode. Outliers in numerical features, such as product dimensions and delivery times, will be addressed using the Interquartile Range (IQR) method to improve data quality and reduce noise in modeling.

Choosing Machine Learning Methods:

1. Customer Segmentation and Behavioral Difference Analysis

This part focuses on understanding behavioral patterns across different groups.

- **Clustering Analysis**:
    - **Methods:** K-Means
    - **Purpose:** To identify natural customer groupings based on **behavioral patterns**, even if we can't directly distinguish between first-time and repeat buyers. For instance, cluster customers based on their customer_state, customer_city, and order-related features like payment type or product categories to see if specific regional or behavioral groups exist and if their cancellation/return rates differ.
- **Descriptive Statistics & Hypothesis Testing**:
    - **Purpose:** Before diving into prediction, use statistical methods to confirm if there are significant differences in order cancellation or product return behaviors across your identified segments.

2. Predictive Models (Identifying High-Risk Customers)

The target of this process is aiming to predict order cancellations or returns.

- Methods:
- **Logistic Regression**: Simple and highly interpretable; good for a baseline model.
- **Random Forests**: Can capture nonlinear relationships. Random Forests, by combining multiple decision trees, are more robust and less prone to overfitting.
- **Support Vector Machines (SVM)**: Perform well on high-dimensional data but can be computationally intensive.
- **Recurrent Neural Network (RNN):** An RNN model will be constructed to capture temporal dependencies and sequential patterns in customer behavior data, especially where browsing or shopping actions occur over time.
    - One input layer (equal to the number of input features after encoding)
    - One or more hidden layers using activation function like ReLU
    - One output layer using a sigmoid activation for binary classification

The RNN will be trained using the Adam optimizer and binary cross-entropy loss.

Model Evaluation:

Performance will be evaluated using accuracy, precision, recall, F1-score, and AUC. Regularization techniques such as dropout and early stopping will be applied to prevent overfitting.

All models will be evaluated based on:

- Precision, Recall, and F1-score, with emphasis on Recall to reduce false negatives
- ROC-AUC to assess class separation quality
- Confusion Matrix for detailed error analysis

<u>References</u>

[1] S. Choi, M.Opitz, Y. Qu, M. Samuel. “EDA”. Arizona State University: DAT 490 Section 45358/45359 online submission, 6/6/2025.